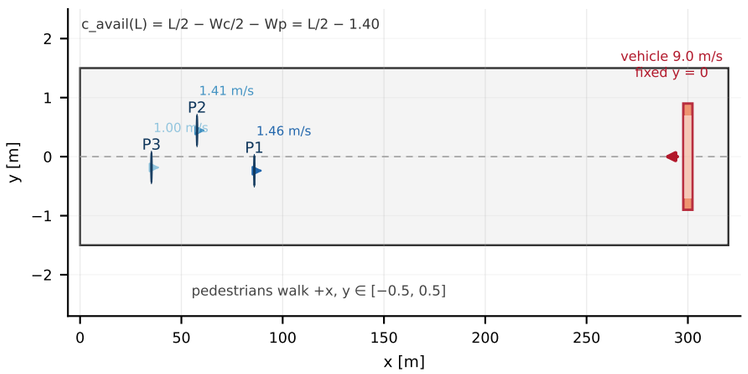
Week 06 — narrow-road pedestrian–vehicle encounter: one unified agent model, two experiments (130 runs)

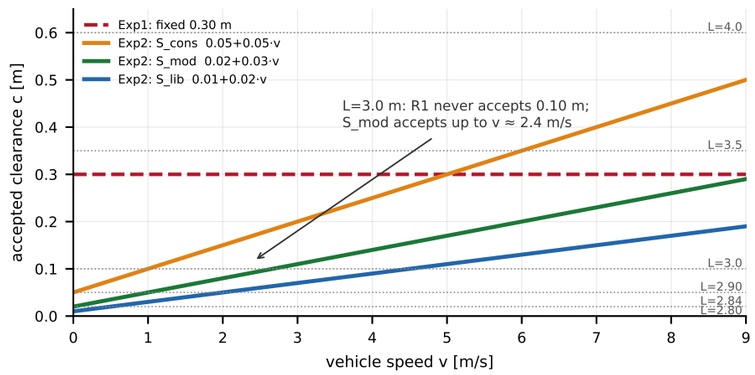
Exp1 (fixed 0.30 m personal space): boundary 3.40 m (R0) / 2.90 m (R1); L=3.0 m is a rule failure. Exp2 (speed-dependent acceptance c_acc(v)=c0+k·v): boundary 2.84 / 2.82 m; delay cut by 40–70%.

social force (SFM) + Wiedemann longitudinal models · deterministic · 55 tests · zero collisions

(a) scenario: narrow mixed road, no sidewalk



(b) pedestrian model: fixed vs speed-dependent acceptance

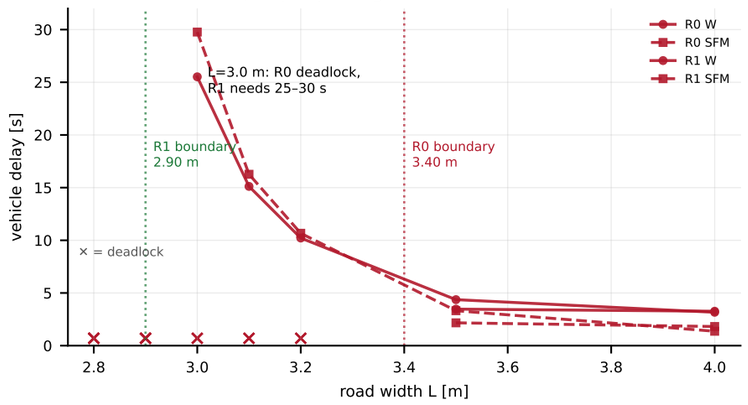


(c) all key results (delay: Wiedemann / SFM)

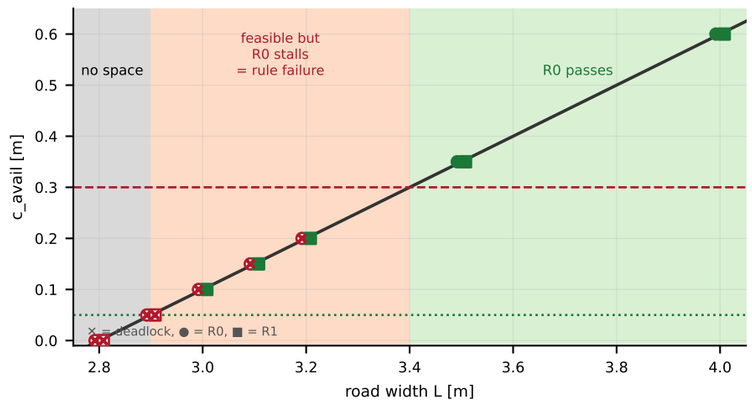
case	result	delay [s]	clear
Exp1 L=4.0 R0	pass	3.3 / 1.8	0.60
Exp1 L=3.0 R0	deadlock (stop 68 s)	—	0.10
Exp1 L=3.0 R1	pass (crawl 1 m/s)	25.5 / 29.8	0.10
Exp1 L=2.8 R1	deadlock: no space	—	0.00
Exp2 L=3.0 S_cons	pass	27.9 / 26.8	0.10
Exp2 L=3.0 S_mod	pass (2.4 m/s)	15.3 / 14.5	0.10
Exp2 L=3.0 S_lib	pass (4.0 m/s)	8.5 / 8.1	0.10
Exp2 L=2.84 S_mod	pass: sequential	33.0 / 31.6	0.02
Exp2 L=2.82 S_lib	pass: sequential	33.0 / 31.6	0.01
Exp2 L=2.80 any	deadlock: geometric limit	—	0.00

boundaries: Exp1 3.40 m (R0) · 2.90 m (R1) Exp2 2.84 m (S_mod) · 2.82 m (S_lib) limit 2.80 m
clear = lateral body clearance at the passing moment

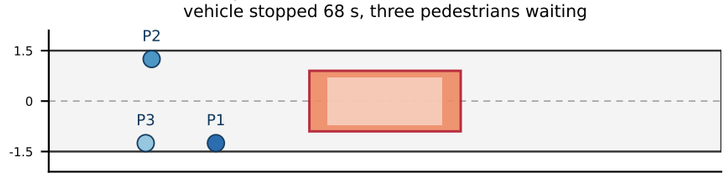
(d) Exp1: passage cost and deadlocks



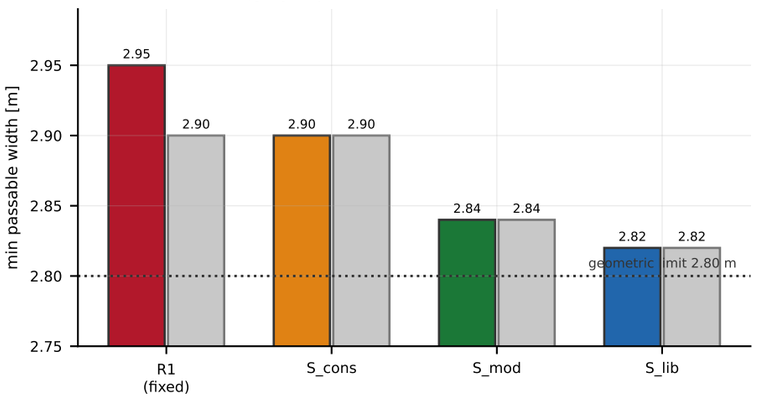
(e) Exp1: space insufficiency vs rule failure



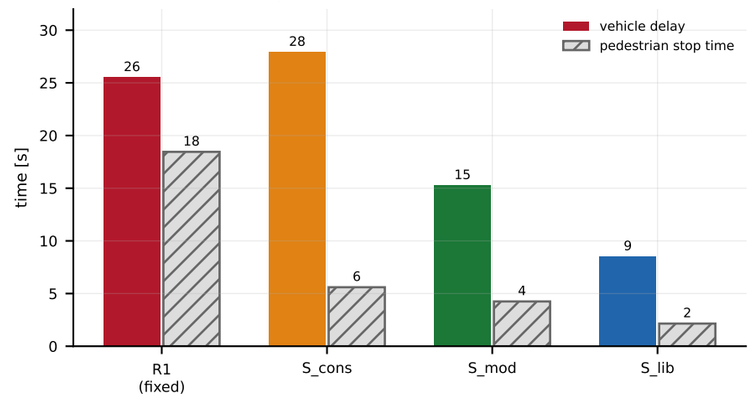
(f) Exp1 L=3.0 m R0: deadlock, standoff 2.5 m
vehicle stopped 68 s, three pedestrians waiting



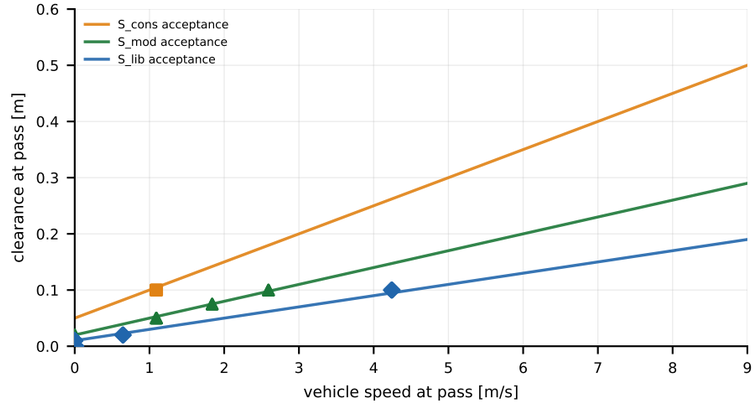
(g) Exp2: boundary 2.90 → 2.84 → 2.82 m
(grey = predicted, colour = simulated)



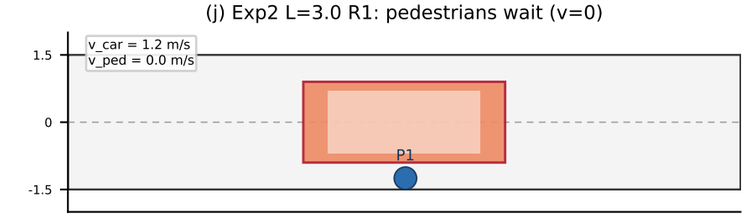
(h) Exp2: cost at L=3.0 m (Wiedemann)



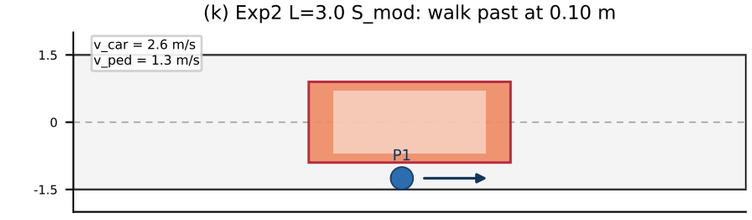
(i) Exp2: measured passing events lie on
the acceptance curves



(j) Exp2 L=3.0 R1: pedestrians wait (v=0)



(k) Exp2 L=3.0 S_mod: walk past at 0.10 m



(l) conclusions

1. R0 (fixed 0.30 m) deadlocks at L=3.0 m although 0.10 m is geometrically available → rule failure, not lack of space. A1 (fix vehicle rule) passes; A2 (fix pedestrian rule) stalls.
2. Fixed personal space for L ≤ 2.90 m is space insufficiency: c_avail ≤ 0.05 m, every rule deadlocks; 2.80 m is the hard limit.
3. Speed-dependent acceptance c_acc(v)=c0+k·v (pedestrians accept small gaps when the car is slow; speed can drop to 0) moves the boundary to 2.84 / 2.82 m and cuts the L=3.0 m vehicle delay from 25–30 s to 15 / 8 s; pedestrian waiting 20 s → 2–4 s.
4. At the boundary the mechanism is sequential: the car stops and waits while pedestrians walk past at 0.01–0.02 m.
5. The decisive element is the decision-level acceptance threshold; SFM force scaling is second order (<0.5 s).
6. 130 runs: zero collisions, no emergency braking (|a| ≤ 2.5 m/s²).

reproduce: cd narrow_road && python run_experiments.py · python run_acceptance_experiment.py · python -m pytest -q